

DEMİR DEMİRDÖĞEN

Software Engineering Student

Izmir, Turkey

 [linkedin/demir-demirdöğen](#)  [demirdogen.dev](#)  [this-demir](#)

Education

Yaşar University

Sep 2023 – Present

Bachelor of Science in Software Engineering | GPA: 3.23 / 4.00

Izmir, Turkey

- **Relevant Coursework:** Data Structures, Algorithms, Object-Oriented Programming (OOP), Database Management Systems, Software Testing.

Engineering Experience

Yu-Sync | [yu-sync.com](#) 

Aug 2025 – Present

Founder & Full Stack Developer

Independent Venture

- Architected a high-performance, bilingual course scheduling web platform serving **7,000+ unique visitors** as a volunteer initiative enabling students to create their timetables.
- Engineered a custom **Bitmasking-based Backtracking algorithm** running on **Web Workers**, enabling conflict detection and schedule generation in milliseconds without blocking the main UI thread.
- Optimized system scalability using Cloudflare, achieving a **92% cache rate** and a **78.7% Click-Through Rate** on Google Search Console.

Selected Projects

Health Journey Mobile SaaS | *React Native, TypeScript, Supabase*

2025

- Developed a cross-platform (iOS/Android) health tourism marketplace connecting patients with clinics, featuring **Real-time** data syncing and Database Triggers via Supabase.
- Built a responsive UI with **Tailwind CSS** supporting dynamic internationalization (i18n) for 5 languages and implemented a smart search algorithm.

Avo Breeze E-Commerce Architecture | *.NET 8, MSSQL, N-Layer Architecture*

2025

- Built a scalable backend system using **.NET 8** and **N-Layer Architecture**; secured API endpoints with **JWT** and optimized data consistency using Entity Framework Core.

Vulkan BVH Ray Tracer | *Java, Vulkan API (LWJGL), GLSL*

2025

- Developed a real-time Ray Tracing engine using **Compute Shaders** and **Bounding Volume Hierarchy (BVH)** to optimize intersection tests, achieving logarithmic complexity for dynamic geometries.
- Designed a **Thread-Safe** architecture isolating Rendering, UI, and Scene logic, implementing explicit memory management to maximize CPU/GPU resource utilization.

QA Automation Framework | *Java, Selenium, JUnit 5, IEEE 829*

May 2025 – Jun 2025

- Led the development of a sustainable test automation structure using **Selenium** and **JUnit 5**, enabling continuous regression testing with minimal effort.
- Ensured engineering quality by documenting all test processes and architectural structure in compliance with **IEEE 829** standards.

Technical Skills

Languages: Java, C/C++, C#, TypeScript, JavaScript, Python, SQL, GLSL

Core Competencies: OOP, Data Structures & Algorithms, Distributed Systems, Multithreading, Memory Management

Technologies & Frameworks: .NET 8, React, React Native, Spring Boot, Vulkan API, Docker, Selenium, JUnit 5

Tools: Git, Postman, Vercel, Supabase, VS Code, JetBrains IDEs, Cloudflare

Languages: Turkish (Native), English (Professional Working Proficiency - B2)

Achievements & Interests

National Athlete (Fencing): Former National Team athlete representing Turkey in international competitions.

DENEYAP Technology Workshops: Completed a 3-year technology training program in robotics and coding.

Interests: Low-level Graphics Programming, Basketball, Online Games.